Lab 21,22,23

M TILAK VENKATA SAI

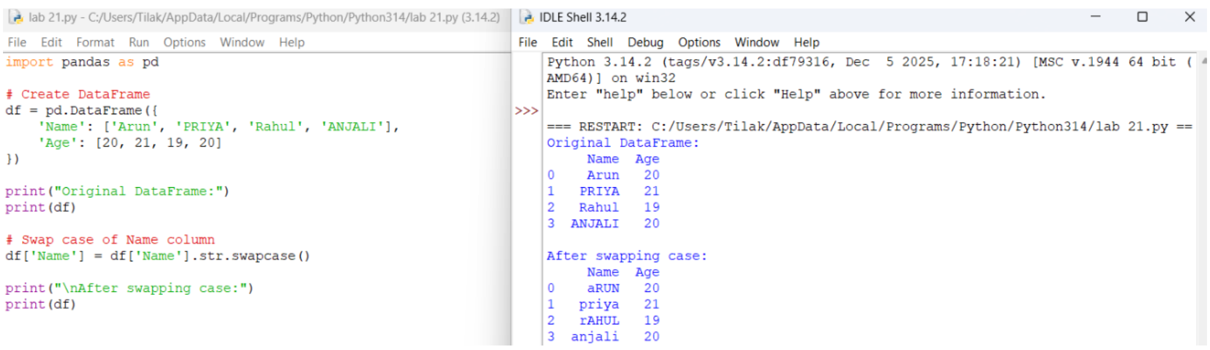
192424415

DSA0504

21. Swap the cases of a specified character column

Code

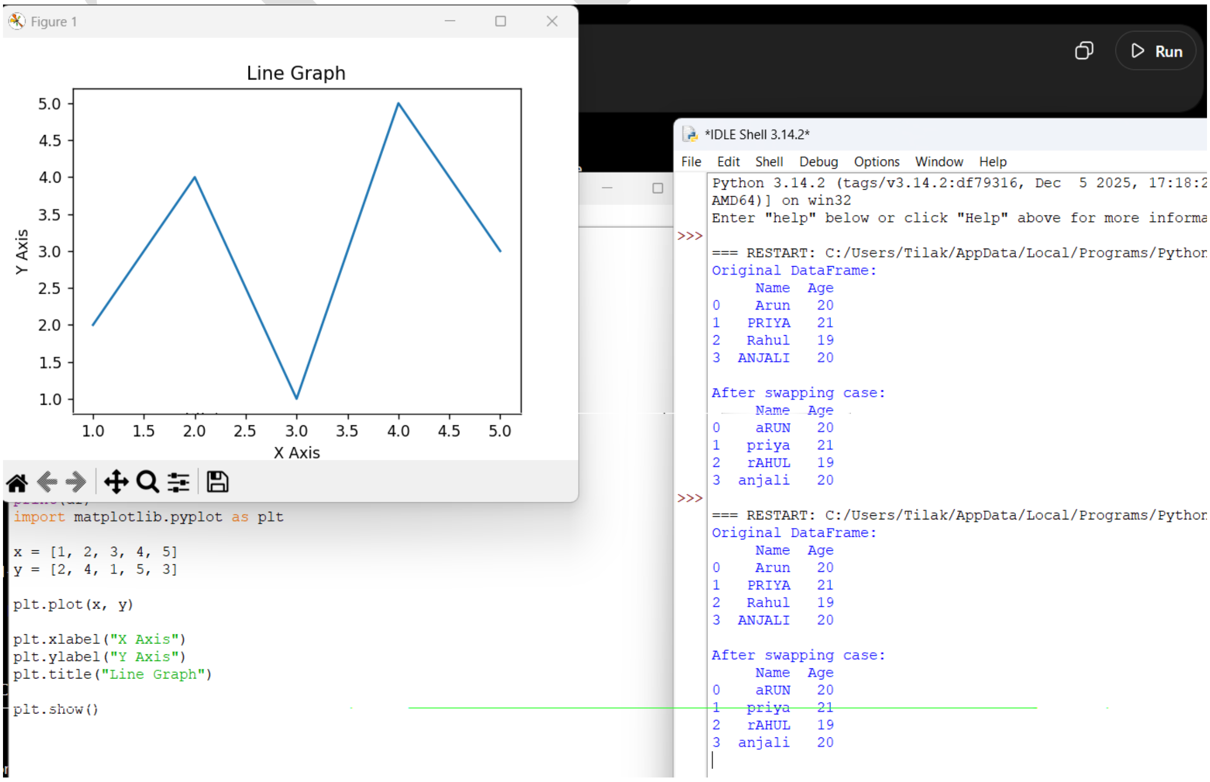
```
import pandas as pd
# Create DataFrame
df = pd.DataFrame({
  'Name': ['Arun', 'PRIYA', 'Rahul', 'ANJALI'],
  'Age': [20, 21, 19, 20]
})

print("Original DataFrame:")
print(df)

# Swap case of Name column
df['Name'] = df['Name'].str.swapcase()

print("\nAfter swapping case:")
print(df)
```

22. Draw a line with suitable X-axis, Y-axis labels and title

Code

```
import matplotlib.pyplot as plt


x = [1, 2, 3, 4, 5]

y = [2, 4, 1, 5, 3]


plt.plot(x, y)


plt.xlabel("X Axis")

plt.ylabel("Y Axis")

plt.title("Line Graph")


plt.show()
```

23. Draw a line using values from a text file

First create a text file called:

test.txt

Put exactly:

```
1 2

2 4

3 1
```

Save it in the **same folder as your Python program**.

Code

```
import matplotlib.pyplot as plt


x = []

y = []


# Read data from text file

with open("test.txt", "r") as file:

  for line in file:

    values = line.split()

    x.append(float(values[0]))

    y.append(float(values[1]))


# Draw line

plt.plot(x, y)


plt.xlabel("X Axis")
```

plt.ylabel("Y Axis")

plt.title("Line Graph from Text File")

plt.show()

Graph values

The file contains:

X	Y
1	2
2	4
3	1